

Advanced AI Dissertation

Quantum Ensemble Stacking with State Preservation

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Abstract

This dissertation investigates a fully quantum stacking architecture for ensemble classification on Noisy Intermediate-Scale Quantum (NISQ) hardware. Classical ensemble stacking requires measuring base learner outputs prior to their transmission to a meta-learner, a process that collapses quantum superposition and discards the amplitude and phase information encoded in the quantum state. A measurement-free alternative is proposed and empirically evaluated, in which multiple small Variational Quantum Circuit (VQC) base learners execute in parallel on separate qubit registers of a single IBM quantum device, with their unmeasured output states routed directly into a quantum meta-learner circuit. Three experimental conditions are evaluated: a classical stacking baseline, a quantum meta-learner operating on re-encoded classical outputs, and the novel measurement-free quantum stacking architecture. All conditions are assessed across established benchmark classification datasets under noiseless simulation, noisy simulation, and real IBM quantum hardware, with Qiskit Runtime error mitigation applied throughout. The central empirical question is whether preserving quantum state information across the stacking boundary yields a measurable classification benefit, and whether that benefit is maintained under NISQ-era noise.

Research Notes: Colour-Coding Key

This chapter is a **temporary reference** to be removed before final submission. It documents the colour-coding system used when annotating quotes and evidence throughout the research and writing process. Each colour corresponds to a category of information; when recording a note, write it in the appropriate colour using the commands below.

Colour Command Definitions:

```
\motive{text}      % Blue   - Motivation
\challenge{text}   % Red    - Challenge
\limitation{text}  % Orange - Limitation
\method{text}      % Cyan   - Method
\result{text}      % Yellow - Result
\benchmark{text}   % Purple - Benchmark
\comparison{text}  % Pink   - Comparison
\theory{text}      % Grey   - Theory
\tooling{text}     % Brown  - Tooling
\gap{text}         % Black  - Gap
\validation{text}  % Green  - Validation
```

Motivation

Argues why something is worth doing. Use for justifying your research questions, the measurement-collapse premise, and the value of ensemble diversity in QML.

Challenge

Identifies a known problem or obstacle. Use for barren plateaus, SWAP overhead, decoherence, training instability, and coherence-time limitations on NISQ hardware.

Limitation

Documents a specific shortcoming of an existing approach. Particularly useful for flagging simulation-only scope, narrow dataset choice, or restricted circuit ansatz.

Method

Describes how something is done. Use for circuit architectures, training procedures, error mitigation techniques, ensemble construction strategies, and decomposition schemes.

Result

Reports concrete empirical findings. Use for accuracy numbers, convergence rates, hardware-versus-simulation gaps, and qubit resource counts.

Benchmark

Establishes a performance baseline or reference point against which other work can be compared. Use for state-of-the-art accuracy figures, circuit depth baselines, and classical ML comparisons.

Comparison

Directly contrasts two or more approaches, models, or configurations. Use when a paper explicitly positions one method against another, or when you are drawing your own cross-paper comparison.

Theory

Provides formal or mathematical grounding. Use for the no-cloning theorem, expressibility proofs, gradient variance analysis, circuit complexity bounds, and quantum information-theoretic arguments.

Tooling

Describes software, frameworks, or hardware. Use for Qiskit, PennyLane, IBM Eagle/Heron, Qiskit Runtime, `mtthree`, and any other concrete implementation resource.

Gap

Explicitly identifies something that has not been studied. These are high-value notes for novelty justification in your introduction and related work chapters.

Validation

Demonstrates that a method works on real hardware or a non-toy dataset. Use for papers that go beyond simulation to confirm results on actual IBM machines or domain-relevant data.

Chapter 1: Introduction

1.1 Motivation

Quantum Machine Learning (QML) occupies an unsettled frontier in which the NISQ era has produced a family of hybrid quantum-classical algorithms, most notably Variational Quantum Circuits (VQCs), whose theoretical promise has yet to be confirmed in practice. A structural limitation shared by virtually all current QML pipelines is their universal application of a measurement step before any aggregation, combination, or further learning takes place. In classical machine learning, reading a neuron's activation leaves its value unchanged; in quantum mechanics, measurement is irreversible, collapsing the superposition state into a definite classical outcome and discarding all phase and amplitude information encoded in that state. This asymmetry between the two computational paradigms motivates the central inquiry of this dissertation: whether, in an ensemble of quantum classifiers, deferring measurement until after the meta-learner has acted on the full quantum state is both feasible and beneficial.

1.2 Problem Statement

Ensemble learning is among the most reliably effective paradigms in classical machine learning, with stacking enabling a meta-learner to exploit the combined outputs of diverse base learners and frequently achieve higher accuracy than any constituent model alone. In quantum computing, ensemble methods have begun to receive attention, yet virtually all proposed architectures follow a measure-then-combine pipeline that is structurally identical to its classical counterpart: base classifiers produce classical probability vectors, which a classical or quantum meta-learner subsequently processes, discarding the quantum nature of the base classifiers at the point of measurement. This work proposes and evaluates a measurement-free stacking architecture, referred to throughout as Condition C, in which base VQC circuits execute in parallel on spatially separated qubit registers of a single IBM quantum processor, with their output states routed directly into a quantum meta-learner circuit without intermediate measurement. The complete classification pipeline, from input encoding through to final prediction, is executed as a single unitary transformation, with measurement occurring only at the terminal stage.

1.3 Research Questions

This dissertation is structured around three specific, falsifiable research questions:

1. Does a measurement-free quantum meta-learner achieve higher classification accuracy than a classical stacking baseline under noiseless simulation?
2. Does that advantage persist under realistic NISQ noise, as modelled by a noisy simulator

and confirmed on IBM hardware?

3. How does a quantum meta-learner operating on re-encoded classical outputs compare to the measurement-free architecture?

1.4 Scope and Contributions

Chapter 2: **Background**

Chapter 3: **Related Work**

Chapter 4: Methodology

4.1 Experimental Conditions

4.2 Circuit Architecture

4.3 Training Strategy

4.4 Simulation Framework

4.5 Hardware Deployment

4.6 Evaluation Metrics

Chapter 5: Results and Analysis

5.1 Noiseless Simulation Results

5.2 Noisy Simulation Results

5.3 IBM Hardware Results

5.4 Discussion

Chapter 6: **Summary and Conclusions**

6.1 Summary of Findings

6.2 Contributions

6.3 Limitations and Future Work

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